

SIMON MILZ – CURRICULUM VITAE

DATE OF BIRTH	May 23, 1987	ADDRESS	David Brewster Building
NATIONALITY	German		Heriot-Watt University
LANGUAGES	German (native), English (fluent), French (proficient)		Edinburgh
EMAIL	s.milzs@hw.ac.uk		Currie EH14 4AS
ORCID	0000-0002-6987-5513		United Kingdom

PROFESSIONAL EXPERIENCE

02/2025 – present	Assistant Professor Theoretical Quantum Information Institute of Photonics and Quantum Sciences (IPAQS) Heriot-Watt University Edinburgh
04/2023 – 02/2025	Marie-Sklodowska-Curie (MSCA) Postdoctoral Fellow Quantum Information Theory Group School of Physics Trinity College Dublin
10/2022 – 04/2023	Co-Group Leader Young Independent Research Group (YIRG) School of Physics University of Vienna
03/2022 – 04/2023	Senior Postdoctoral Fellow Young Independent Research Group (YIRG) Institute for Quantum Optics and Quantum Information (IQOQI) Vienna, Austrian Academy of Sciences
12/2019 – 03/2022	Postdoctoral Fellow/ ESQ Fellow Huber Group Institute for Quantum Optics and Quantum Information (IQOQI) Vienna Austrian Academy of Sciences
05/2019 – 12/2019	Postdoctoral Fellow YIRG (Young Independent Research Group) Institute for Quantum Optics and Quantum Information (IQOQI) Vienna Austrian Academy of Sciences

EDUCATION

11/2015 – 04/2019	PhD in Quantum Information Theory Quantum Information Science Group Monash University, Melbourne, Australia. THESIS: On the Operational Theory of General (Quantum) Stochastic Processes SUPERVISORS: Kavan Modi and Felix A. Pollock EXAMINERS: Howard M. Wiseman and Paolo Perinotti
06/2018 – 09/2018	Research Visit. Department of Computer Science University of Oxford, UK TOPIC: Resource Theory of Causal Connection COLLABORATION with Giulio Chiribella
04/2011 – 08/2011	Internship at the Institute of Theoretical Physics Université Aix-Marseille II, France. TOPIC: Pseudo-Anosov homeomorphisms in hyperelliptic components SUPERVISOR: Erwan Lanneau

- 06/2010 – 09/2011 **Study exchange.**
 Université Aix-Marseille II, France.
 Masters programme for Mathematical and Theoretical Physics
- 10/2007 – 04/2014 **Diplom in Physics*** (GPA: 1.1/1.0)**.
 Theoretical Quantum Optics Group
 TU Dresden, Germany
- THESIS: Correlations and their Influence on the Dynamics of Open Qubit Systems.
 SUPERVISOR: Walter T. Strunz
- (*Equivalent to a MSc in Physics)
 (** German academic grades range from 1.0 to 5.0, with 1.0 being best.)

FELLOWSHIPS AND AWARDS

- 04/2023 – PRESENT **Marie Skłodowska-Curie (MSCA) Postdoctoral Fellowship** (~ 200k EUR)
 Trinity College Dublin, Ireland
- 2019 – 2021 **ESQ Fellowship** (~ 120k EUR)
 Erwin Schrödinger Center for Quantum Science & Technology (ESQ), Austria
- 2019 **Monash Postgraduate Publication Award** (~3.2k EUR)
 Funding for high-achieving PhD students for preparation and publication of research results after graduation
- 2015 – 2019 **J. L. William Postgraduate Scholarship** (~ 9.4k EUR),
 Top-up Scholarship offered by the School of Physics and Astronomy,
 Monash University, Melbourne
- 2015 – 2019 **Monash Graduate Scholarship** (MGS) (~ 63k EUR)
 Stipend and Tuition Scholarship for students undertaking Research Doctorate degrees
- 2010 – 2014 **Studienstiftung des Deutschen Volkes**
 (German Academic Scholarship Foundation),
 Scholarship foundation funded by the German government for the support of academically exceptional students

EXTERNAL FUNDING

- 2019 – 2021 **Foundational Questions Institute (FQXi)**, Large Grant (~ 68k EUR),
 TITLE: Predictive Quantum Intelligence under Physical Constraints
- 2018 – 2020 **Australia-Germany joint Research co-operation scheme**, (~ 26k EUR)
 Travel grant by the German Academic Exchange Service (DAAD)
 Awarded to instigate collaboration between the Theoretical Quantum Optics Group at the University Siegen, Germany, and the Quantum Information Science Group at Monash University, Australia
 TITLE: Genuinely quantum stochastic processes

SUPERVISION AND MENTORING ACTIVITIES

- 07/2024 – present **PhD co-supervision** of Yutong Luo
 Trinity College Dublin
 PROJECT TITLE: Athermality, memory, and the arrow of time in quantum mechanics.
- 09/2023 – 07/2024 **MSc co-supervision** of Yutong Luo
 Trinity College Dublin
 PROJECT TITLE: The role of memory in the resource theory of athermality

- 2023 **Capstone project (9 weeks) co-supervision** of Minfei Zou and Dubhaltach Mc-Sweeny
Trinity College Dublin
Project titles: Efficient approximations to stochastic process models & Petz recovery of multi-time processes
- 2020 – 2022 **PhD co-supervision** of Philip Taranto
University of Vienna
Thesis: Quantum Information Processing: Thermodynamics, Complexity, and Multi-Time Phenomena
- 2020 – 2021 **MSc co-supervision** of Felix Hubmann
University of Vienna
Project title: Thermal operations with memory
- Since 2020 **Thesis advisory committee (TAC)** mentor of Joshua Morris
University of Vienna
- 2018 – 2019 **MSc co-supervision** of Philip Taranto
Monash University
Thesis: An Introduction to Memory Effects in Quantum Processes
- 2018 **Honours co-supervision** of Dominic Jurkschat
Monash University
Thesis: Causality in Quantum Processes.

TEACHING ACTIVITIES

- SINCE 01/2024 **Co-Lecturer** on graduate course SELECTED TOPICS IN QUANTUM INFORMATION THEORY
Coordination, Design and Teaching
Trinity College Dublin
- 10/2020 – 04/2021 **Co-Lecturer** on graduate course ENTANGLEMENT THEORY
Coordination, Design, Teaching, and Marking
University of Vienna
- 11/2015 – 01/2019 **Tutorial classes** on Quantum Mechanics, Thermodynamics, and Electrodynamics
Preparation and delivery of tutorials for undergraduate students
School of Physics & Astronomy, Monash University, Melbourne, Australia
- 2011 – 2013 **Student research assistant**
Theoretical Quantum Optics Group, TU Dresden, Germany.
Preparation of exercise and solution sheets for student tutorials
- 2008 – 2010 **Tutorial classes on Analysis**
Institute for Analysis, TU Dresden, Germany.
Preparation and delivery of tutorials for undergraduates

REVIEWING ACTIVITIES AND ACADEMIC CITIZENSHIP

Scientific referee for:

- Nature Communications
- Physical Review X
- Physical Review Letters
- Physical Review A
- npj Quantum Information
- Quantum
- New Journal of Physics

Organizer of the Physics Colloquium at Monash University (2016-2018)

Peer-reviewed in International Journals

1. CHARACTERISING THE HIERARCHY OF MULTI-TIME QUANTUM PROCESSES WITH CLASSICAL MEMORY
P. Taranto, M. T. Quintino, M. Murao, and [SM](#)
[Quantum 8, 1328 \(2024\)](#).
2. TRANSFORMATIONS BETWEEN ARBITRARY (QUANTUM) OBJECTS AND THE EMERGENCE OF INDEFINITE CAUSALITY
[SM](#) and M. T. Quintino
[Quantum 8, 1328 \(2024\)](#).
3. WITNESSING ENVIRONMENT DIMENSION THROUGH TEMPORAL CORRELATIONS
L. B. Vieira, [SM](#), G. Vitagliano, and C. Budroni
[Quantum 8, 1224 \(2024\)](#).
4. HIDDEN QUANTUM MEMORY: IS MEMORY THERE WHEN SOMEBODY LOOKS?
P. Taranto, Thomas J. Elliott, and [SM](#)
[Quantum 7, 991 \(2023\)](#).
5. EXTRACTING QUANTUM DYNAMICAL RESOURCES: CONSUMPTION OF NON-MARKOVIANITY FOR NOISE REDUCTION
G. D. Berk, [SM](#), F. A. Pollock, and K. Modi
[npj Quantum Inf 9, 104 \(2023\)](#).
6. RESOURCE THEORY OF CAUSAL CONNECTION
[SM](#), J. Bavaresco, and G. Chiribella
[Quantum 6, 788 \(2022\)](#).
7. CONNECTING COMMUTATIVITY AND CLASSICALITY FOR MULTI-TIME QUANTUM PROCESSES
F. Sakuldee, P. Taranto, and [SM](#)
[Phys. Rev. A 106, 022416 \(2022\)](#).
8. *Quantum stochastic processes and quantum non-Markovian phenomena*
[SM](#) and K. Modi,
[PRX Quantum 2, 030201 \(2021\)](#).
9. GENUINE MULTIPARTITE ENTANGLEMENT IN TIME
[SM](#), C. Spee, Z.-P. Xu, F. A. Pollock, K. Modi, and O. Gühne,
[SciPost Phys. 10, 141 \(2021\)](#).
10. DELAYED-CHOICE CAUSAL ORDER AND NONCLASSICAL CORRELATIONS
[SM](#), D. Jurkschat, F. A. Pollock, and K. Modi,
[Phys. Rev. Research 3, 023028 \(2021\)](#).
11. WHEN IS A NON-MARKOVIAN QUANTUM PROCESS CLASSICAL?
[SM](#), D. Egloff, P. Taranto, T. Theurer, M. B. Plenio, A. Smirne, and S. F. Huelga,
[Phys. Rev. X 10, 041049 \(2020\)](#).
12. KOLMOGOROV EXTENSION THEOREM FOR (QUANTUM) CAUSAL MODELLING AND GENERAL PROBABILISTIC THEORIES
[SM](#), F. Sakuldee, F. A. Pollock, and K. Modi,
[Quantum 4, 255 \(2020\)](#).
13. COMPLETELY POSITIVE DIVISIBILITY DOES NOT MEAN MARKOVIANITY
[SM](#), M. S. Kim, F. A. Pollock, and K. Modi,
[Phys. Rev. Lett. 123, 040401 \(2019\)](#).
14. QUANTUM MARKOV ORDER
P. Taranto, F. A. Pollock, [SM](#), M. Tomamichel, and K. Modi,
[Phys. Rev. Lett. 122, 140401 \(2019\)](#).
15. STRUCTURE OF QUANTUM STOCHASTIC PROCESSES WITH FINITE MARKOV ORDER
P. Taranto, [SM](#), F. A. Pollock, and K. Modi,
[Phys. Rev. A 99, 042108 \(2019\)](#).
16. RECONSTRUCTING OPEN QUANTUM SYSTEM DYNAMICS WITH LIMITED CONTROL
[SM](#), F. A. Pollock, and K. Modi,
[Phys. Rev. A, 98, 012108 \(2018\)](#).

17. NON-MARKOVIAN QUANTUM CONTROL AS COHERENT STOCHASTIC TRAJECTORIES
F. Sakuldee, [SM](#), F. A. Pollock, and K. Modi,
[J. Phys. A: Math. Theor.](#) **51**, 414014 (2018).
18. ENTANGLEMENT, NON-MARKOVIANITY, AND CAUSAL NON-SEPARABILITY
[SM](#), F. A. Pollock, T. P. Le, G. Chiribella, and K. Modi,
[New J. Phys.](#) **20**, 033033 (2018).
19. AN INTRODUCTION TO OPERATIONAL QUANTUM DYNAMICS
[SM](#), F. A. Pollock, and K. Modi,
[Open Syst. Inf. Dyn.](#) **24**, 1740016 (2017).
20. VOLUMES OF CONDITIONED BIPARTITE STATE SPACES
[SM](#) and W. T. Strunz
[J. Phys. A: Math. Theor.](#) **48**, 035306 (2014).

Pre-Prints

1. THERMODYNAMIC CRITERIA FOR SIGNALING IN QUANTUM CHANNELS
Yutong Luo, [SM](#), Felix C. Binder
[arXiv:2506.20428](#) (2025).
2. HIGHER-ORDER QUANTUM OPERATIONS
P. Taranto, [SM](#), M. Murao, M. T. Quintino, and K. Modi
[arXiv:2503.09693](#) (2025).
3. MONOTONES IN RESOURCE THEORIES FOR DYNAMICAL DECOUPLING
G. D. Berk, [SM](#), and K. Modi
[arXiv:2412.11595](#) (2024).